

Competitive Programming Notebook

Programadores Roblox

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1 Search and sort

1.1 Monotonic Stack

```

1 vector<int> find_esq(vector<int> &v, bool maior) {
2     int n = v.size();
3     vector<int> result(n);
4     stack<int> s;
5
6     for (int i = 0; i < n; i++) {
7         while (!s.empty() && (maior ? v[s.top()] <= v[i] : v[
8             s.top()] >= v[i])) {
9             s.pop();
10        }
11        if (s.empty()) {
12            result[i] = -1;
13        } else {
14            result[i] = v[s.top()];
15        }
16        s.push(i);
17    }
18    return result;
19 }
20
21 vector<int> find_dir(vector<int> &v, bool maior) {
22     int n = v.size();
23     vector<int> result(n);
24     stack<int> s;
25     for (int i = n - 1; i >= 0; i--) {
26         while (!s.empty() && (maior ? v[s.top()] <= v[i] : v[
27             s.top()] >= v[i])) {
28             s.pop();
29        }
30        if (s.empty()) {
31            result[i] = -1;
32        } else {
33            result[i] = v[s.top()];
34        }
35        s.push(i);
36    }
37    return result;
38 }

```

1.2 Merge Sort

```

1 int mergeAndCount(vector<int>& v, int l, int m, int r) {
2     int x = m - l + 1;
3     int y = r - m;
4     vector<int> left(x), right(y);
5     for (int i = 0; i < x; i++) left[i] = v[l + i];
6     for (int j = 0; j < y; j++) right[j] = v[m + 1 + j];
7     int i = 0, j = 0, k = l;
8     int swaps = 0;
9     while (i < x && j < y) {
10        if (left[i] <= right[j]) {
11            v[k++] = left[i++];
12        } else {
13            v[k++] = right[j++];
14            swaps += (x - i);
15        }
16    }
17    while (i < x) v[k++] = left[i++];
18    while (j < y) v[k++] = right[j++];
19    return swaps;
20 }
21
22 int mergeSort(vector<int>& v, int l, int r) {
23     int swaps = 0;
24     if (l < r) {
25         int m = l + (r - l) / 2;
26         swaps += mergeSort(v, l, m);
27         swaps += mergeSort(v, m + 1, r);
28         swaps += mergeAndCount(v, l, m, r);
29     }
30     return swaps;
31 }

```

2 Stress

2.1 Gen

```

1 // pre-compilar os headers:
2 // compilar template com g++ -H e procurar onde esta stdc++.h
3 // criar a pasta bits e incluir com "" ou inves de <
4 // compilar stress com: g++ -pipe -O3 -flto -march=native -
5 // mtune=native gen.cpp
6 // faz bastante diferena no runtime
7
8 #include <bits/stdc++.h>
9 #include <cstdlib>
10 #include <ctime>
11 using namespace std;
12
13 int randi(int L, int R) { return L + rand() % (R - L + 1); }
14 char randc(char L, char R) { return char(L + rand() % (R - L
15     + 1)); }
16
17 int main(int argc, char** argv) {
18     if (argc > 1) srand(atoi(argv[1]));
19     else srand(time(0));
20
21     int n = randi(1, 100);
22     cout << n << '\n';
23     for (int i = 0; i < n; i++) {
24         cout << randi(-10, 20) << ' ';
25     }
26     cout << '\n';
27 }
28
29 /*
30 script.sh
31
32 g++ -std=c++17 -O2 -o sol sol.cpp
33 g++ -std=c++17 -O2 -o brute brute.cpp
34 g++ -std=c++17 -O2 -o gen gen.cpp
35
36 for ((i = 1; ; i++)); do
37     ./gen $i > in.txt
38     ./sol < in.txt > out1.txt
39     ./brute < in.txt > out2.txt
40
41     if ! diff -q out1.txt out2.txt > /dev/null; then
42         echo "Mismatch found on test $i"
43         cat in.txt
44         echo "Your output:"
45         cat out1.txt
46         echo "Brute output:"
47         cat out2.txt
48         break
49     fi
50     echo "Test $i passed"
51 done
52
53 */

```

3 Geometry

3.1 Lattice Points

```

1 ll gcd(ll a, ll b) {
2     return b == 0 ? a : gcd(b, a % b);
3 }
4
5 ll area_triangulo(ll x1, ll y1, ll x2, ll y2, ll x3, ll y3) {
6     return abs(x1 * (y2 - y3) + x2 * (y3 - y1) + x3 * (y1 -
7         y2));
8 }
9
10 ll pontos_borda(ll x1, ll y1, ll x2, ll y2) {
11     return gcd(abs(x2 - x1), abs(y2 - y1));
12 }
13
14 int32_t main() {
15     ll x1, y1, x2, y2, x3, y3;
16     cin >> x1 >> y1;
17     cin >> x2 >> y2;
18     cin >> x3 >> y3;
19     ll area = area_triangulo(x1, y1, x2, y2, x3, y3);
20     ll tot_borda = pontos_borda(x1, y1, x2, y2) +
21         pontos_borda(x2, y2, x3, y3) + pontos_borda(x3, y3, x1,
22             y1);
23
24     ll ans = (area - tot_borda) / 2 + 1;
25     cout << ans << endl;
26
27     return 0;
28 }

```

3.2 Point Location

```

1
2 int32_t main(){
3     sws;
4
5     int t; cin >> t;
6
7     while(t--){
8
9         int x1, y1, x2, y2, x3, y3; cin >> x1 >> y1 >> x2 >>
            y2 >> x3 >> y3;
10
11         int deltax1 = (x1-x2), deltax1 = (y1-y2);
12
13         int compx = (x1-x3), compy = (y1-y3);
14
15         int ans = (deltax1*compy) - (compx*deltax1);
16
17         if(ans == 0){cout << "TOUCH\n"; continue;}
18         if(ans < 0){cout << "RIGHT\n"; continue;}
19         if(ans > 0){cout << "LEFT\n"; continue;}
20     }
21     return 0;
22 }

```

3.3 Convex Hull

```

1 #include <bits/stdc++.h>
2
3 using namespace std;
4 #define int long long
5 typedef int cod;
6
7 struct point
8 {
9     cod x,y;
10    point(cod x = 0, cod y = 0): x(x), y(y)
11    {}
12
13    double modulo()
14    {
15        return sqrt(x*x + y*y);
16    }
17
18    point operator+(point o)
19    {
20        return point(x+o.x, y+o.y);
21    }
22    point operator-(point o)
23    {
24        return point(x - o.x, y - o.y);
25    }
26    point operator*(cod t)
27    {
28        return point(x*t, y*t);
29    }
30    point operator/(cod t)
31    {
32        return point(x/t, y/t);
33    }
34
35    cod operator*(point o)
36    {
37        return x*o.x + y*o.y;
38    }
39    cod operator^(point o)
40    {
41        return x*o.y - y * o.x;
42    }
43    bool operator<(point o)
44    {
45        if( x != o.x) return x < o.x;
46        return y < o.y;
47    }
48
49 };
50
51 int ccw(point p1, point p2, point p3)
52 {
53     cod cross = (p2-p1) ^ (p3-p1);
54     if(cross == 0) return 0;
55     else if(cross < 0) return -1;
56     else return 1;
57 }

```

```

58
59 vector <point> convex_hull(vector<point> p)
60 {
61     sort(p.begin(), p.end());
62     vector<point> L,U;
63
64     //Lower
65     for(auto pp : p)
66     {
67         while(L.size() >= 2 and ccw(L[L.size()-2], L.back(),
68             , pp) == -1)
69         {
70             // ÃT -1 pq eu nÃo quero excluir os colineares
71             L.pop_back();
72         }
73         L.push_back(pp);
74     }
75     reverse(p.begin(), p.end());
76
77     //Upper
78     for(auto pp : p)
79     {
80         while(U.size() >= 2 and ccw(U[U.size()-2], U.back(),
81             pp) == -1)
82         {
83             U.pop_back();
84         }
85         U.push_back(pp);
86     }
87     L.pop_back();
88     L.insert(L.end(), U.begin(), U.end()-1);
89     return L;
90 }
91
92 cod area(vector<point> v)
93 {
94     int ans = 0;
95     int aux = (int)v.size();
96     for(int i = 2; i < aux; i++)
97     {
98         ans += ((v[i] - v[0])^(v[i-1] - v[0]))/2;
99     }
100    ans = abs(ans);
101    return ans;
102 }
103
104 int bound(point p1, point p2)
105 {
106     return __gcd(abs(p1.x-p2.x), abs(p1.y-p2.y));
107 }
108 //teorema de pick [pontos = A - (bound+points)/2 + 1]
109
110 int32_t main()
111 {
112
113     int n;
114     cin >> n;
115
116     vector<point> v(n);
117     for(int i = 0; i < n; i++)
118     {
119         cin >> v[i].x >> v[i].y;
120     }
121
122     vector <point> ch = convex_hull(v);
123
124     cout << ch.size() << '\n';
125     for(auto p : ch) cout << p.x << " " << p.y << "\n";
126
127     return 0;
128 }

```

3.4 Point

```

1 template<class T>
2 struct Point {
3     typedef Point P;
4     T x, y;
5     explicit Point(T x=0, T y=0) : x(x), y(y) {}
6     bool operator<(P p) const { return tie(x,y) < tie(p.x,p.y)
7     }; }
8     bool operator==(P p) const { return tie(x,y)==tie(p.x,p.y)
9     }; }

```

```

8   P operator+(P p) const { return P(x+p.x, y+p.y); }
9   P operator-(P p) const { return P(x-p.x, y-p.y); }
10  P operator*(T d) const { return P(x*d, y*d); }
11  P operator/(T d) const { return P(x/d, y/d); }
12  T operator^(P p) const { return x*p.y - y*p.x; }
13  T dot(P p) const { return x*p.x + y*p.y; } // ||a|| * ||b||
14  T cross(P p) const { return x*p.y - y*p.x; } // ||a|| * ||b|| * sin(theta)
15  T cross(P a, P b) const { return (a-*this).cross(b-*this); }
16  T dist2() const { return x*x + y*y; }
17  double dist() const { return sqrt((double)dist2()); }
18
19  T orient(P a, P b, P c) {return (b-a)^(c-a);}
20
21  // angle to x-axis in interval [-pi, pi]
22  double angle() const { return atan2(y, x); }
23  P unit() const { return *this/dist(); } // makes dist()=1
24  P perp() const { return P(-y, x); } // rotates +90 degrees
25  P normal() const { return perp().unit(); }
26  // returns point rotated 'a' radians ccw around the origin
27  P rotate(double a) const {
28      return P(x*cos(a)-y*sin(a), x*sin(a)+y*cos(a));
29  }
30
31  friend ostream& operator<<(ostream& os, P p) {
32      return os << "(" << p.x << ", " << p.y << ")";
33  }
34
35  // inside the disk of diameter [a, b]
36  bool inDisk(P a, P b, P p) {
37      return (a-p).dot(b-p) <= 0;
38  }
39
40  bool onSegment(P a, P b, P p) {
41      return orient(a,b,p) == 0 && inDisk(a,b,p);
42  }
43
44  bool properInter(P a, P b, P c, P d, P &ans) {
45      double oa = orient(c,d,a),
46             ob = orient(c,d,b),
47             oc = orient(a,b,c),
48             od = orient(a,b,d);
49      // Proper intersection exists iff opposite signs
50      if (oa*ob < 0 && oc*od < 0) {
51          P ans = (a*ob - b*oa) / (ob-oa);
52          return true;
53      }
54      return false;
55  }
56
57  set<P> inters(P a, P b, P c, P d) {
58      P out;
59      if (properInter(a,b,c,d,out)) return {out};
60      set<P> s;
61      if (onSegment(c,d,a)) s.insert(a);
62      if (onSegment(c,d,b)) s.insert(b);
63      if (onSegment(a,b,c)) s.insert(c);
64      if (onSegment(a,b,d)) s.insert(d);
65      return s;
66  }
67 };

```

3.5 Inside Polygon

```

1  // Convex O(logn)
2
3  bool insideT(point a, point b, point c, point e){
4      int x = ccw(a, b, e);
5      int y = ccw(b, c, e);
6      int z = ccw(c, a, e);
7      return !((x==1 or y==1 or z==1) and (x==-1 or y==-1 or z==-1));
8  }
9
10 bool inside(vp &p, point e){ // ccw
11     int l=2, r=(int)p.size()-1;
12     while(l<r){
13         int mid = (l+r)/2;
14         if(ccw(p[0], p[mid], e) == 1)
15             l=mid+1;
16         else{
17             r=mid;

```

```

18     }
19 }
20 // bordo
21 // if(r==(int)p.size()-1 and ccw(p[0], p[r], e)==0)
22     return false;
23 // if(r==2 and ccw(p[0], p[1], e)==0) return false;
24 // if(ccw(p[r], p[r-1], e)==0) return false;
25 return insideT(p[0], p[r-1], p[r], e);
26 }
27
28 // Any O(n)
29
30 int inside(vp &p, point pp){
31     // 1 - inside / 0 - boundary / -1 - outside
32     int n = p.size();
33     for(int i=0; i<n; i++){
34         int j = (i+1)%n;
35         if(line({p[i], p[j]}).inside_seg(pp))
36             return 0;
37     }
38     int inter = 0;
39     for(int i=0; i<n; i++){
40         int j = (i+1)%n;
41         if(p[i].x <= pp.x and pp.x < p[j].x and ccw(p[i], p[j], pp)==1)
42             inter++; // up
43         else if(p[j].x <= pp.x and pp.x < p[i].x and ccw(p[i], p[j], pp)==-1)
44             inter++; // down
45     }
46
47     if(inter%2==0) return -1; // outside
48     else return 1; // inside
49 }

```

4 DS

4.1 Bit

```

1 struct BIT {
2     int n;
3     vector<int> bit;
4     BIT(int n = 0): n(n), bit(n + 1, 0) {}
5     void add(int i, int delta) {
6         for(; i <= n; i += i & -i) bit[i] += delta;
7     }
8     int sum(int i) {
9         int r = 0;
10        for(; i > 0; i -= i & -i) r += bit[i];
11        return r;
12    }
13    int range_sum(int l, int r){
14        if (r < 1) return 0;
15        return sum(r) - sum(l - 1);
16    }
17 };

```

4.2 Sparse Table

```

1 // O-index, O(1)
2 struct SparseTable {
3     vector<vector<int>>> st;
4     int max_log;
5     SparseTable(vector<int>& arr) {
6         int n = arr.size();
7         max_log = floor(log2(n)) + 1;
8         st.resize(n, vector<int>(max_log));
9         for (int i = 0; i < n; i++) {
10             st[i][0] = arr[i];
11         }
12         for (int j = 1; j < max_log; j++) {
13             for (int i = 0; i + (1 << j) <= n; i++) {
14                 st[i][j] = max(st[i][j - 1], st[i + (1 << (j - 1))][j - 1]);
15             }
16         }
17     }
18     int query(int L, int R) {
19         int tamanho = R - L + 1;
20         int k = floor(log2(tamanho));
21         return max(st[L][k], st[R - (1 << k) + 1][k]);
22     }
23 };

```

4.3 Min Queue

```

1 struct MinQueue {
2     deque<pair<int, int>> dq;
3     void push(int val, int idx) {
4         while (!dq.empty() && dq.back().first >= val) dq.
5             pop_back();
6         dq.emplace_back(val, idx);
7     }
8     void pop(int idx) {
9         if (!dq.empty() && dq.front().second == idx) {
10             dq.pop_front();
11         }
12     }
13     int get() {
14         return dq.front().first;
15     }
16     bool empty() {
17         return dq.empty();
18     }
19 };
20 /*
21 considerando janela de tamanho K
22 mq.push(v[i], i);
23 if (i >= k) {
24     mq.pop(i - k);
25 }
26 if (i >= k - 1) {
27     result.push_back(mq.get());
28 }
29 */

```

4.4 Segtree Iterativa

```

1 // Exemplo de uso:
2 // SegTree<int> st(vetor);
3 // range query e point update
4 template <typename T>
5 struct SegTree {
6     int n;
7     vector<T> tree;
8     T neutral_value = 0;
9     T combine(T a, T b) {
10         return a + b;
11     }
12
13     SegTree(const vector<T>& data) {
14         n = data.size();
15         tree.resize(2 * n, neutral_value);
16
17         for (int i = 0; i < n; i++)
18             tree[n + i] = data[i];
19
20         for (int i = n - 1; i > 0; --i)
21             tree[i] = combine(tree[i * 2], tree[i * 2 + 1]);
22     }
23
24     T query(int l, int r) {
25         T res_l = neutral_value, res_r = neutral_value;
26
27         for (l += n, r += n + 1; l < r; l >>= 1, r >>= 1) {
28             if (l & 1) res_l = combine(res_l, tree[l++]);
29             if (r & 1) res_r = combine(tree[--r], res_r);
30         }
31
32         return combine(res_l, res_r);
33     }
34
35     void update(int pos, T new_val) {
36         tree[pos += n] = new_val;
37         for (pos >>= 1; pos > 0; pos >>= 1)
38             tree[pos] = combine(tree[2 * pos], tree[2 * pos + 1]);
39     }
40 };

```

4.5 Merge Sort Tree

```

1 #define all(x) x.begin(), x.end()
2
3 struct MST {
4     int n;
5     vector<vector<int>> tree;
6     MST(vector<int> &a) {
7         n = a.size();
8         tree.resize(4 * n);

```

```

9         build(1, 0, n - 1, a);
10     }
11
12     void build(int x, int lx, int rx, vector<int> &a) {
13         if (lx == rx) {
14             tree[x] = { a[lx] };
15             return;
16         }
17         int mid = lx + (rx - lx) / 2;
18         build(2 * x, lx, mid, a);
19         build(2 * x + 1, mid + 1, rx, a);
20         auto &L = tree[2 * x], &R = tree[2 * x + 1];
21         tree[x].resize(L.size() + R.size());
22         merge(all(L), all(R), tree[x].begin());
23     }
24
25     int query(int x, int lx, int rx, int l, int r, int val) {
26         if (lx > r || rx < l) return 0;
27         if (lx >= l && rx <= r) {
28             auto &v = tree[x];
29             return lower_bound(all(v), val) - v.begin();
30         }
31         int mid = lx + (rx - lx) / 2;
32         return query(2 * x, lx, mid, l, r, val) + query(2 * x + 1, mid + 1, rx, l, r, val);
33     }
34
35     int query(int l, int r, int val) {
36         if (l > r) return 0;
37         return query(1, 0, n - 1, l, r, val);
38     }
39 };
40
41 /* mst.query(l, r, l) retorna quantos caras distintos no
42     range
43     map<int, int> last;
44     for (int i = 0; i < n; i++) {
45         if (last.count(a[i])) {
46             esq[i] = last[a[i]];
47         } else {
48             esq[i] = -1;
49         }
50         last[a[i]] = i;
51     }
52     MST mst(esq);
53     jogar vetor de ultima aparicao na seg
54 */

```

4.6 Seglazy Iterativa

```

1 template <typename T, typename L>
2 struct SegLazy {
3     int n, h;
4     vector<T> tree;
5     vector<L> lazy;
6     vector<bool> has_lazy;
7     vector<int> sz; // se len = pow2, remove e sz[p] = 1 << h
8     [p]
9
10     // --- Mudar aqui ---
11     const T T_neutral = LLONG_MAX;
12
13     inline T combine(T a, T b) {
14         return min(a, b);
15     }
16
17     inline T apply_lazy_to_node(T node, L delta, int len) {
18         return node + (T)delta;
19     }
20
21     inline L combine_lazy(L old_delta, L new_delta) {
22         return old_delta + new_delta;
23     }
24
25     // -----
26
27     SegLazy(const vector<T>& data) {
28         n = data.size();
29         h = 32 - __builtin_clz(n);
30         tree.assign(2 * n, T_neutral);
31         lazy.assign(n, L());
32         has_lazy.assign(n, false);
33         sz.assign(2 * n, 1);
34
35         for (int i = 0; i < n; i++) tree[n + i] = data[i];
36         for (int i = n - 1; i > 0; --i) {
37             tree[i] = combine(tree[i << 1], tree[i << 1 | 1]);
38
39             ;
40             sz[i] = sz[i << 1] + sz[i << 1 | 1];
41         }

```

```

38 }
39
40 inline void apply(int p, L v) {
41     tree[p] = apply_lazy_to_node(tree[p], v, sz[p]);
42     if (p < n) {
43         if (has_lazy[p]) lazy[p] = combine_lazy(lazy[p],
44 v);
45         else lazy[p] = v;
46         has_lazy[p] = true;
47     }
48 }
49
50 inline void build(int p) {
51     p += n;
52     while (p > 1) {
53         p >>= 1;
54         T res = combine(tree[p << 1], tree[p << 1 | 1]);
55         if (has_lazy[p]) tree[p] = apply_lazy_to_node(res,
56 lazy[p], sz[p]);
57         else tree[p] = res;
58     }
59 }
60
61 inline void push(int p) {
62     p += n;
63     for (int s = h; s > 0; --s) {
64         int i = p >> s;
65         if (has_lazy[i]) {
66             apply(i << 1, lazy[i]);
67             apply(i << 1 | 1, lazy[i]);
68             has_lazy[i] = false;
69         }
70     }
71 }
72
73 void update(int l, int r, L v) {
74     int l0 = l, r0 = r;
75     push(l); push(r);
76     for (l += n, r += n + 1; l < r; l >>= 1, r >>= 1) {
77         if (l & 1) apply(l++, v);
78         if (r & 1) apply(--r, v);
79     }
80     build(l0); build(r0);
81 }
82
83 T query(int l, int r) {
84     push(l); push(r);
85     T res_l = T_neutral, res_r = T_neutral;
86     for (l += n, r += n + 1; l < r; l >>= 1, r >>= 1) {
87         if (l & 1) res_l = combine(res_l, tree[l++]);
88         if (r & 1) res_r = combine(tree[--r], res_r);
89     }
90     return combine(res_l, res_r);
91 }
92
93 };

```

4.7 Psum 2d

```

1 vector<vector<int>>> psum(h+1, vector<int>(w+1, 0));
2 for (int i=1; i<=h; i++){
3     for (int j=1; j<=w; j++){
4         cin >> psum[i][j];
5         psum[i][j] += psum[i-1][j]+psum[i][j-1]-psum[i-1][j
6         -1];
7     }
8 }
9 // retorna a psum2d do intervalo inclusivo [(a, b), (c, d)]
10 int retangulo(int a, int b, int c, int d){
11     c = min(c, h), d = min(d, w);
12     a = max(0LL, a-1), b = max(0LL, b-1);
13     return v[c][d]-v[a][d]-v[c][b]+v[a][b];
14 }

```

4.8 Bit2d

```

1 // 1-index
2 struct BIT2d {
3     int n, m;
4     vector<vector<int>>> tree;
5     BIT2d(int n, int m) : n(n), m(m) {
6         tree.assign(n + 1, vector<int>(m + 1, 0));
7     }
8     void add(int y, int x, int delta) {
9         for (int i = y; i <= n; i += i & -i) {

```

```

10         for (int j = x; j <= m; j += j & -j) {
11             tree[i][j] += delta;
12         }
13     }
14 }
15
16 int pref(int y, int x) {
17     int res = 0;
18     for (int i = y; i > 0; i -= i & -i) {
19         for (int j = x; j > 0; j -= j & -j) {
20             res += tree[i][j];
21         }
22     }
23     return res;
24 }
25
26 int query(int y1, int x1, int y2, int x2) {
27     return pref(y2, x2) - pref(y1 - 1, x2) - pref(y2, x1 - 1)
28     + pref(y1 - 1, x1 - 1);
29 }

```

4.9 Segtree Lazy

```

1 /*
2 Seg com double Lazy de soma e de set
3 tipo 1 - soma em range
4 tipo 2 - set em range
5 */
6 struct SegTree {
7     int n;
8     vector<int> v, tree;
9     vector<int> lazy_soma, lazy_set;
10     SegTree(vector<int> &a) : v(a), n(a.size()) {
11         tree.resize(4 * n);
12         lazy_soma.resize(4 * n, 0);
13         lazy_set.resize(4 * n, -1);
14         build(1, 0, n - 1);
15     };
16
17 void build(int x, int lx, int rx) {
18     if (lx == rx) {
19         tree[x] = v[lx];
20         return;
21     }
22     int mid = (lx + rx) / 2;
23     build(2 * x, lx, mid);
24     build(2 * x + 1, mid + 1, rx);
25     tree[x] = tree[2 * x] + tree[2 * x + 1];
26 }
27
28 void seta(int x, int lx, int rx, int val) {
29     tree[x] = val * (rx - lx + 1);
30     lazy_set[x] = val;
31     lazy_soma[x] = 0;
32 }
33
34 void soma(int x, int lx, int rx, int val) {
35     tree[x] += val * (rx - lx + 1);
36     if (lazy_set[x] != -1) {
37         lazy_set[x] += val;
38     } else {
39         lazy_soma[x] += val;
40     }
41 }
42
43 void push(int x, int lx, int rx) {
44     int mid = (lx + rx) / 2;
45     if (lazy_set[x] != -1) {
46         if (lx != rx) {
47             seta(2 * x, lx, mid, lazy_set[x]);
48             seta(2 * x + 1, mid + 1, rx, lazy_set[x]);
49         }
50         lazy_set[x] = -1;
51     }
52     if (lazy_soma[x] != 0) {
53         if (lx != rx) {
54             soma(2 * x, lx, mid, lazy_soma[x]);
55             soma(2 * x + 1, mid + 1, rx, lazy_soma[x]);
56         }
57         lazy_soma[x] = 0;
58     }
59 }
60
61 void update(int x, int lx, int rx, int l, int r, int val,
62 int type) {
63     push(x, lx, rx);
64     if (lx >= l && rx <= r) {
65         if (type == 1) soma(x, lx, rx, val);
66         else seta(x, lx, rx, val);
67         push(x, lx, rx);
68         return;
69     }

```

```

64     if (lx > r || rx < l) return;
65     int mid = (lx + rx) / 2;
66     update(2 * x, lx, mid, l, r, val, type);
67     update(2 * x + 1, mid + 1, rx, l, r, val, type);
68     tree[x] = tree[2 * x] + tree[2 * x + 1];
69 }
70 void update(int l, int r, int val, int type) {
71     update(1, 0, n - 1, l, r, val, type);
72 }
73 int query(int x, int lx, int rx, int l, int r) {
74     push(x, lx, rx);
75     if (lx >= l && rx <= r) return tree[x];
76     if (lx > r || rx < l) return 0;
77     int mid = (lx + rx) / 2;
78     return query(2 * x, lx, mid, l, r) + query(2 * x + 1, mid
79         + 1, rx, l, r);
80 }
81 int query(int l, int r) {
82     return query(1, 0, n - 1, l, r);
83 }
84 };

```

4.10 Seg Lazy Pa

```

1  /* Notas
2  PA eh da forma a0 (a) e razão d)
3  na hora de propagar o lazy
4  aplica no no S = (a0 + (a0 + (n - 1) * d)) * n / 2 (
5  variante da soma total do no)
6  pros filhos, o da esquerda eh normal
7  lazy_a[esq] += lazy_a[x]
8  lazy_d[esq] += lazy_d[x]
9  pro da direita tem que mudar o a (que eh o elemento inicial
10  naquele no da direita)
11  lazy_a[dir] += a + len_esq * (d)
12  lazy_d[dir] += lazy_d[x]
13 */
14 int gauss(int n, int a, int d) {
15     // (a0 + an) * n / 2
16     if (n <= 0)
17         return 0;
18     return (a + (a + (n - 1) * d)) * n / 2;
19 }
20 struct SegTree {
21     int n;
22     vector<int> v, lazy_a, lazy_d, tree;
23     SegTree(vector<int> &a) : v(a), n(a.size()) {
24         tree.resize(4 * n);
25         lazy_a.resize(4 * n, 0);
26         lazy_d.resize(4 * n, 0);
27         build(1, 0, n - 1);
28     }
29     void build(int x, int lx, int rx) {
30         if (lx == rx) {
31             tree[x] = v[lx];
32             return;
33         }
34         int mid = (lx + rx) / 2;
35         build(2 * x, lx, mid);
36         build(2 * x + 1, mid + 1, rx);
37         tree[x] = tree[2 * x] + tree[2 * x + 1];
38     }
39     int query(int x, int lx, int rx, int l, int r) {
40         push(x, lx, rx);
41         if (lx >= l && rx <= r)
42             return tree[x];
43         if (lx > r || rx < l)
44             return 0;
45         int mid = (lx + rx) / 2;
46         return query(2 * x, lx, mid, l, r) + query(2 * x + 1, mid
47             + 1, rx, l, r);
48     }
49     int query(int l, int r) {
50         return query(1, 0, n - 1, l, r);
51     }
52     void push(int x, int lx, int rx) {
53         if (lazy_a[x] && lazy_d[x]) {
54             int tam = rx - lx + 1;
55             tree[x] += gauss(tam, lazy_a[x], lazy_d[x]);
56             if (lx != rx) {
57                 int mid = (lx + rx) / 2;
58                 int tam_esq = mid - lx + 1;
59                 lazy_a[2 * x] += lazy_a[x];

```

```

59                 lazy_d[2 * x] += lazy_d[x];
60                 lazy_a[2 * x + 1] += (lazy_a[x] + tam_esq * lazy_d[x]
61                     );
62                 lazy_d[2 * x + 1] += lazy_d[x];
63             }
64             lazy_a[x] = lazy_d[x] = 0;
65         }
66     }
67     void update(int x, int lx, int rx, int l, int r, int a, int
68         d) {
69         push(x, lx, rx);
70         if (lx >= l && rx <= r) {
71             lazy_a[x] += a + (lx - l) * d;
72             lazy_d[x] += d;
73             push(x, lx, rx);
74             return;
75         }
76         if (lx > r || rx < l)
77             return;
78         int mid = (lx + rx) / 2;
79         update(2 * x, lx, mid, l, r, a, d);
80         update(2 * x + 1, mid + 1, rx, l, r, a, d);
81         tree[x] = tree[2 * x] + tree[2 * x + 1];
82     }
83     void update(int l, int r, int a, int d) {
84         update(1, 0, n - 1, l, r, a, d);
85     }
86 };

```

4.11 Trie Bits

```

1  int trie[MAXN][2];
2  int cnt[MAXN][2];
3  int cur_node = 0;
4
5  void insere(int x) {
6      int node = 0;
7      for (int i = 29; ~i; i--) {
8          int bit = (1ll << i) & x ? 1 : 0;
9          if (trie[node][bit] == 0) trie[node][bit] = ++
10             cur_node;
11             cnt[node][bit]++;
12             node = trie[node][bit];
13         }
14     }
15     int query(int x) {
16         int r = 0;
17         int node = 0;
18         for (int i = 29; ~i; i--) {
19             int bit = (x >> i) & 1;
20             if (trie[node][bit ^ 1] != 0 && cnt[node][bit ^ 1] >
21                 0) {
22                 r |= (1ll << i);
23                 node = trie[node][bit ^ 1];
24             } else {
25                 node = trie[node][bit];
26             }
27         }
28         return r;
29     }
30 }

```

4.12 Segtree2d

```

1  // 0 index
2  struct SegTree2d {
3      int n;
4      vector<vector<int>> tree;
5      SegTree2d(int n) : n(n) {
6          tree.assign(4 * n, vector<int>(4 * n, 0));
7      }
8      void update_x(int no_y, int no_x, int lx, int rx, int x,
9          int val, bool folha_y) {
10         if (lx == rx) {
11             if (folha_y) tree[no_y][no_x] = val;
12             else tree[no_y][no_x] = tree[2 * no_y][no_x] + tree[2 *
13                 no_y + 1][no_x];
14             return;
15         }
16         int mid = (lx + rx) / 2;
17         if (x <= mid) update_x(no_y, 2 * no_x, lx, mid, x, val,
18             folha_y);
19         else update_x(no_y, 2 * no_x + 1, mid + 1, rx, x, val,
20             folha_y);

```

```

17     tree[no_y][no_x] = tree[no_y][2 * no_x] + tree[no_y][2 *
18         no_x + 1];
19 void update_y(int no_y, int ly, int ry, int y, int x, int
20     val) {
21     if (ly == ry) {
22         update_x(no_y, 1, 0, n - 1, x, val, true);
23         return;
24     }
25     int mid = (ly + ry) / 2;
26     if (y <= mid) update_y(2 * no_y, ly, mid, y, x, val);
27     else update_y(2 * no_y + 1, mid + 1, ry, y, x, val);
28     update_x(no_y, 1, 0, n - 1, x, val, false);
29 }
30 int query_x(int no_y, int no_x, int lx, int rx, int x1, int
31     x2) {
32     if (rx < x1 || lx > x2) return 0;
33     if (lx >= x1 && rx <= x2) return tree[no_y][no_x];
34     int mid = (lx + rx) / 2;
35     return query_x(no_y, 2 * no_x, lx, mid, x1, x2) + query_x
36         (no_y, 2 * no_x + 1, mid + 1, rx, x1, x2);
37 }
38 int query_y(int no_y, int ly, int ry, int y1, int y2, int
39     x1, int x2) {
40     if (ry < y1 || ly > y2) return 0;
41     if (ly >= y1 && ry <= y2) return query_x(no_y, 1, 0, n -
42         1, x1, x2);
43     int mid = (ly + ry) / 2;
44     return query_y(2 * no_y, ly, mid, y1, y2, x1, x2) +
45         query_y(2 * no_y + 1, mid + 1, ry, y1, y2, x1, x2);
46 }
47 void update(int y, int x, int val) {
48     update_y(1, 0, n - 1, y, x, val);
49 }
50 int query(int y1, int x1, int y2, int x2) {
51     return query_y(1, 0, n - 1, y1, y2, x1, x2);
52 }
53 };

```

4.13 Dsu

```

1 struct DSU {
2     vector<int> parent, size;
3     int C;
4     DSU(int n) : parent(n + 1), size(n + 1), C(n) {
5         for (int i = 1; i <= n; i++) {
6             parent[i] = i;
7             size[i] = 1;
8         }
9     }
10    int find(int a) {
11        if (parent[a] == a) return a;
12        return parent[a] = find(parent[a]);
13    }
14    int same(int a, int b) {
15        return find(a) == find(b);
16    }
17    int get_size(int a) {
18        return size[find(a)];
19    }
20    int count() {
21        return C;
22    }
23    void join(int a, int b) {
24        int pA = find(a);
25        int pB = find(b);
26        if (pA != pB) {
27            C--;
28            if (size[pA] > size[pB]) {
29                swap(pA, pB);
30            }
31            parent[pA] = pB;
32            size[pB] += size[pA];
33        }
34    }
35 };

```

4.14 Ordered Set E Map

```

1 #include<ext/pb_ds/assoc_container.hpp>
2 #include<ext/pb_ds/tree_policy.hpp>
3 using namespace __gnu_pbds;
4 using namespace std;

```

```

6 template<typename T> using ordered_multiset = tree<T,
7     null_type, less_equal<T>, rb_tree_tag,
8     tree_order_statistics_node_update>;
9 template<typename T> using o_set = tree<T, null_type, less<T>,
10     rb_tree_tag, tree_order_statistics_node_update>;
11 template<typename T, typename R> using o_map = tree<T, R,
12     less<T>, rb_tree_tag, tree_order_statistics_node_update>;
13
14 int main() {
15     int i, j, k, n, m;
16     o_set<int> st;
17     st.insert(1);
18     st.insert(2);
19     cout << *st.find_by_order(0) << endl; // k-esimo elemento
20     cout << st.order_of_key(2) << endl; // numero de elementos
21         menores que k
22     o_map<int, int> mp;
23     mp.insert({1, 10});
24     mp.insert({2, 20});
25     cout << mp.find_by_order(0)->second << endl; // k-esimo
26         elemento
27     cout << mp.order_of_key(2) << endl; // numero de elementos
28         (chave) menores que k
29     return 0;
30 }

```

4.15 Maximum Subarray Sum Range

```

1 struct SegTree{
2     int n;
3     vector<array<int, 4>> tree;
4     vector<int> v;
5     SegTree(vector<int> &a) : v(a), n(a.size()) {
6         tree.resize(4 * n);
7         build(1, 0, n - 1);
8     }
9     void build(int x, int lx, int rx) {
10        if (lx == rx) {
11            tree[x] = { v[lx], v[lx], v[lx], v[lx] };
12            return;
13        }
14        int mid = lx + (rx - lx) / 2;
15        build(2 * x, lx, mid);
16        build(2 * x + 1, mid + 1, rx);
17        tree[x] = combine(tree[2 * x], tree[2 * x + 1]);
18    }
19    array<int, 4> query(int x, int lx, int rx, int l, int r) {
20        if (lx >= l && rx <= r)
21            return tree[x];
22        if (lx > r || rx < l)
23            return { 0, LINF, LINF, LINF };
24        int mid = lx + (rx - lx) / 2;
25        return combine(query(2 * x, lx, mid, l, r), query(2 * x +
26            1, mid + 1, rx, l, r));
27    };
28    int query(int l, int r) {
29        return max(0ll, query(1, 0, n - 1, l, r)[3]);
30    }
31    void update(int x, int lx, int rx, int pos, int val) {
32        if (lx == rx) {
33            tree[x] = { val, val, val, val };
34            return;
35        }
36        int mid = lx + (rx - lx) / 2;
37        if (pos <= mid)
38            update(2 * x, lx, mid, pos, val);
39        else
40            update(2 * x + 1, mid + 1, rx, pos, val);
41        tree[x] = combine(tree[2 * x], tree[2 * x + 1]);
42    }
43    void update(int pos, int val) {
44        update(1, 0, n - 1, pos, val);
45    }
46    array<int, 4> combine(array<int, 4> x, array<int, 4> y) {
47        return {
48            x[0] + y[0],
49            max(x[1], x[0] + y[1]),
50            max(y[2], y[0] + x[2]),
51            max({x[3], y[3], x[2] + y[1]})
52        };
53    }
54 };

```

5 General

5.1 Mex

```

1 struct MEX {
2     map<int, int> f;
3     set<int> falta;
4     int tam;
5     MEX(int n) : tam(n) {
6         for (int i = 0; i <= n; i++) falta.insert(i);
7     }
8     void add(int x) {
9         f[x]++;
10        if (f[x] == 1 && x >= 0 && x <= tam) {
11            falta.erase(x);
12        }
13    }
14    void rem(int x) {
15        if (f.count(x) && f[x] > 0) {
16            f[x]--;
17            if (f[x] == 0 && x >= 0 && x <= tam) {
18                falta.insert(x);
19            }
20        }
21    }
22    int get() {
23        if (falta.empty()) return tam + 1;
24        return *falta.begin();
25    }
26 };

```

5.2 Bitwise

```

1 int check_kth_bit(int x, int k) {
2     return (x >> k) & 1;
3 }
4
5 void print_on_bits(int x) {
6     for (int k = 0; k < 32; k++) {
7         if (check_kth_bit(x, k)) {
8             cout << k << ' ';
9         }
10    }
11    cout << '\n';
12 }
13
14 int count_on_bits(int x) {
15     int ans = 0;
16     for (int k = 0; k < 32; k++) {
17         if (check_kth_bit(x, k)) {
18             ans++;
19         }
20    }
21    return ans;
22 }
23
24 bool is_even(int x) {
25     return ((x & 1) == 0);
26 }
27
28 int set_kth_bit(int x, int k) {
29     return x | (1 << k);
30 }
31
32 int unset_kth_bit(int x, int k) {
33     return x & ~(1 << k);
34 }
35
36 int toggle_kth_bit(int x, int k) {
37     return x ^ (1 << k);
38 }
39
40 bool check_power_of_2(int x) {
41     return count_on_bits(x) == 1;
42 }

```

5.3 Debug

```

1 template<typename T, typename U>
2 ostream& operator<<(ostream& os, const pair<T, U>& p) {
3     return os << "(" << p.first << ", " << p.second << ")";
4 }
5

```

```

6 template<typename T, typename = decltype(begin(declval<T>()))
7     , typename = enable_if_t<!is_same_v<T, string>>>
8     ostream& operator<<(ostream& os, const T& v) {
9         os << "{";
10        string sep;
11        for (const auto& x : v) os << sep << x, sep = ", ";
12        return os << "}";
13 }
14
15 #define db(x...) cout << "[" << #x << "]: ", [(auto... $){
16     ((cout << $ << "; ", ...); }(x), cout << '\n'

```

5.4 Count Permutations

```

1 // Returns the number of distinct permutations
2 // that are lexicographically less than the string t
3 // using the provided frequency (freq) of the characters
4 // 0(n*freq.size())
5 int countPermLess(vector<int> freq, const string &t) {
6     int n = t.size();
7     int ans = 0;
8
9     vector<int> fact(n + 1, 1), invfact(n + 1, 1);
10    for (int i = 1; i <= n; i++)
11        fact[i] = (fact[i - 1] * i) % MOD;
12    invfact[n] = fexp(fact[n], MOD - 2, MOD);
13    for (int i = n - 1; i >= 0; i--)
14        invfact[i] = (invfact[i + 1] * (i + 1)) % MOD;
15
16    // For each position in t, try placing a letter smaller
17    // than t[i] that is in freq
18    for (int i = 0; i < n; i++) {
19        for (char c = 'a'; c < t[i]; c++) {
20            if (freq[c - 'a'] > 0) {
21                freq[c - 'a']--;
22                int ways = fact[n - i - 1];
23                for (int f : freq)
24                    ways = (ways * invfact[f]) % MOD;
25                ans = (ans + ways) % MOD;
26                freq[c - 'a']++;
27            }
28            if (freq[t[i] - 'a'] == 0) break;
29            freq[t[i] - 'a']--;
30        }
31    }
32    return ans;
33 }

```

5.5 Brute Choose

```

1 vector<int> elements;
2 int N, K;
3 vector<int> comb;
4
5 void brute_choose(int i) {
6     if (comb.size() == K) {
7         for (int j = 0; j < comb.size(); j++) {
8             cout << comb[j] << ' ';
9         }
10        cout << '\n';
11        return;
12    }
13    if (i == N) return;
14    int r = N - i;
15    int preciso = K - comb.size();
16    if (r < preciso) return;
17    comb.push_back(elements[i]);
18    brute_choose(i + 1);
19    comb.pop_back();
20    brute_choose(i + 1);
21 }

```

5.6 Struct

```

1 struct Pessoa{
2     // Atributos
3     string nome;
4     int idade;
5
6     // Comparador
7     bool operator<(const Pessoa& other) const{
8         if(idade != other.idade) return idade > other.idade;
9         else return nome > other.nome;
10    }
11 }

```

6 String

6.1 Trie

```

1 // Trie por array
2 int trie[MAXN][26];
3 int tot_nos = 0;
4 vector<bool> acaba(MAXN, false);
5 vector<int> contador(MAXN, 0);
6
7 void insere(string s) {
8     int no = 0;
9     for(auto &c : s) {
10         if(trie[no][c - 'a'] == 0) {
11             trie[no][c - 'a'] = ++tot_nos;
12         }
13         no = trie[no][c - 'a'];
14         contador[no]++;
15     }
16     acaba[no] = true;
17 }
18
19 bool busca(string s) {
20     int no = 0;
21     for(auto &c : s) {
22         if(trie[no][c - 'a'] == 0) {
23             return false;
24         }
25         no = trie[no][c - 'a'];
26     }
27     return acaba[no];
28 }
29
30 int isPref(string s) {
31     int no = 0;
32     for(auto &c : s) {
33         if(trie[no][c - 'a'] == 0) {
34             return -1;
35         }
36         no = trie[no][c - 'a'];
37     }
38     return contador[no];
39 }

```

6.2 Z Function

```

1 vector<int> z_function(string s) {
2     int n = s.size();
3     vector<int> z(n);
4     int l = 0, r = 0;
5     for(int i = 1; i < n; i++) {
6         if(i < r) {
7             z[i] = min(r - i, z[i - l]);
8         }
9         while(i + z[i] < n && s[z[i]] == s[i + z[i]]) {
10             z[i]++;
11         }
12         if(i + z[i] > r) {
13             l = i;
14             r = i + z[i];
15         }
16     }
17     return z;
18 }

```

6.3 Kmp

```

1 vector<int> kmp(string &s) {
2     int m = s.size();
3     vector<int> lsp(m, 0);
4     for (int i = 1, j = 0; i < m; i++) {
5         while (j > 0 && s[j] != s[i]) j = lsp[j - 1];
6         if (s[i] == s[j]) j++;
7         lsp[i] = j;
8     }
9     return lsp;
10 }

```

6.4 Hashing

```

1 // String Hash template
2 // constructor(s) - 0(|s|)

```

```

3 // query(l, r) - returns the hash of the range [l,r] from
4 // left to right - 0(1)
5 // query_inv(l, r) from right to left - 0(1)
6 // patrocinado por tiagodfs
7 mt19937 rng(time(nullptr));
8
9 struct Hash {
10     const int X = rng();
11     const int MOD = 1e9+7;
12     int n; string s;
13     vector<int> h, hi, p;
14     Hash() {}
15     Hash(string s): s(s), n(s.size()), h(n), hi(n), p(n) {
16         for (int i=0;i<n;i++) p[i] = (i ? X*p[i-1]:1) % MOD;
17         for (int i=0;i<n;i++)
18             h[i] = (s[i] + (i ? h[i-1]:0) * X) % MOD;
19         for (int i=n-1;i>=0;i--)
20             hi[i] = (s[i] + (i+1<n ? hi[i+1]:0) * X) % MOD;
21     }
22     int query(int l, int r) {
23         int hash = (h[r] - (l ? h[l-1]*p[r-l+1]%MOD : 0));
24         return hash < 0 ? hash + MOD : hash;
25     }
26     int query_inv(int l, int r) {
27         int hash = (hi[l] - (r+1 < n ? hi[r+1]*p[r-l+1] % MOD
28             : 0));
29         return hash < 0 ? hash + MOD : hash;
30     }
31 };

```

7 Math

7.1 Soma Pg Mod

```

1 // 1 + r + r^2 + r^3 + ... r^n (mod m)
2 // 0(log(N)) funciona com qualquer m
3 // retorna {soma, r^n mod m}
4 array<int, 2> soma_pg(int r, int n, int m) {
5     if (n == 0) return {0, 1};
6
7     auto [mid, p] = soma_pg(r, n / 2, m);
8
9     int sum = (mid*p % m + mid) % m;
10    int np = p*p % m; // r^n
11
12    if (n&1) {
13        sum = (sum*r + 1) % m;
14        np = np*r % m;
15    }
16
17    return {sum, np};
18 }

```

7.2 Fexp

```

1 // a^e mod m
2 // 0(log n)
3
4 int fexp(int a, int e, int m) {
5     a %= m;
6     int ans = 1;
7     while (e > 0) {
8         if (e & 1) ans = ans*a % m;
9         a = a*a % m;
10        e /= 2;
11    }
12    return ans%m;
13 }

```

7.3 Base Calc

```

1 int char_to_val(char c) {
2     if (c >= '0' && c <= '9') return c - '0';
3     else return c - 'A' + 10;
4 }
5
6 char val_to_char(int val) {
7     if (val >= 0 && val <= 9) return val + '0';
8     else return val - 10 + 'A';
9 }
10

```

```

11 int to_base_10(string &num, int bfrom) {
12     int result = 0;
13     int pot = 1;
14     for (int i = num.size() - 1; i >= 0; i--) {
15         if (char_to_val(num[i]) >= bfrom) return -1;
16         result += char_to_val(num[i]) * pot;
17         pot *= bfrom;
18     }
19     return result;
20 }
21
22 string from_base_10(int n, int bto) {
23     if (n == 0) return "0";
24     string result = "";
25     while (n > 0) {
26         result += val_to_char(n % bto);
27         n /= bto;
28     }
29     reverse(result.begin(), result.end());
30     return result;
31 }
32
33 string convert_base(string &num, int bfrom, int bto) {
34     int n_base_10 = to_base_10(num, bfrom);
35     return from_base_10(n_base_10, bto);
36 }

```

7.4 Totient

```

1 // phi(n) = n * (1 - 1/p1) * (1 - 1/p2) * ...
2 int phi(int n) {
3     int result = n;
4     for (int i = 2; i * i <= n; i++) {
5         if (n % i == 0) {
6             while (n % i == 0)
7                 n /= i;
8             result -= result / i;
9         }
10    }
11    if (n > 1) // SE n sobrou, ele Ã um fator primo
12        result -= result / n;
13    return result;
14 }
15
16 // crivo phi
17 const int MAXN_PHI = 1000001;
18 int phiv[MAXN_PHI];
19 void phi_sieve() {
20     for (int i = 0; i < MAXN_PHI; i++) phiv[i] = i;
21     for (int i = 2; i < MAXN_PHI; i++) {
22         if (phiv[i] == i) {
23             for (int j = i; j < MAXN_PHI; j += i) phiv[j] -=
24                 phiv[j] / i;
25         }
26     }
27 }

```

7.5 Equacao Diofantina

```

1 int extended_gcd(int a, int b, int& x, int& y) {
2     if (a == 0) {
3         x = 0;
4         y = 1;
5         return b;
6     }
7     int x1, y1;
8     int gcd = extended_gcd(b % a, a, x1, y1);
9     x = y1 - (b / a) * x1;
10    y = x1;
11    return gcd;
12 }
13
14 bool solve(int a, int b, int c, int& x0, int& y0) {
15     int x, y;
16     int g = extended_gcd(abs(a), abs(b), x, y);
17     if (c % g != 0) {
18         return false;
19     }
20     x0 = x * (c / g);
21     y0 = y * (c / g);
22     if (a < 0) x0 = -x0;
23     if (b < 0) y0 = -y0;
24     return true;
25 }

```

7.6 Menor Fator Primo

```

1 const int MAXN = 2e6 + 2;
2 int spf[MAXN];
3 vector<int> primes;
4
5 void sieve() {
6     for (int i = 0; i < MAXN; i++) spf[i] = i;
7     for (int i = 2; i * i < MAXN; i++) {
8         if (spf[i] == i) {
9             for (int j = i * i; j < MAXN; j += i) {
10                 if (spf[j] == j) {
11                     spf[j] = i;
12                 }
13             }
14         }
15     }
16     for (int i = 2; i < MAXN; i++) {
17         if (spf[i] == i) {
18             primes.push_back(i);
19         }
20     }
21 }
22
23 map<int, int> factorize(int n) {
24     map<int, int> factors;
25     while (n > 1) {
26         factors[spf[n]]++;
27         n /= spf[n];
28     }
29     return factors;
30 }
31
32 int numero_de_divisores(int n) {
33     if (n == 1) return 1;
34     map<int, int> fatores = fatorar(n);
35     int nod = 1;
36     for (auto &[primo, expoente] : fatores) nod *= (expoente
37         + 1);
38     return nod;
39 }
40
41 // DEFINE INT LONG LONG
42 int soma_dos_divisores(int n) {
43     if (n == 1) return 1;
44     map<int, int> fatores = fatorar(n);
45     int sod = 1;
46     for (auto &[primo, expoente] : fatores) {
47         int termo_soma = 1;
48         int potencia_primo = 1;
49         for (int i = 0; i < expoente; i++) {
50             potencia_primo *= primo;
51             termo_soma += potencia_primo;
52         }
53         sod *= termo_soma;
54     }
55     return sod;
56 }

```

7.7 Exgcd

```

1 // 0 retorno da funcao eh {n, m, g}
2 // e significa que gcd(a, b) = g e
3 // n e m sao inteiros tais que an + bm = g
4 array<ll, 3> exgcd(int a, int b) {
5     if (b == 0) return {1, 0, a};
6     auto [m, n, g] = exgcd(b, a % b);
7     return {n, m - a / b * n, g};
8 }

```

7.8 Crivo

```

1 // O(n*log(log(n)))
2 bool composto[MAX]
3 for (int i = 1; i <= n; i++) {
4     if (composto[i]) continue;
5     for (int j = 2*i; j <= n; j += i)
6         composto[j] = 1;
7 }

```

7.9 Matrix

```

1  const int MOD = 1e9 + 7;
2
3  struct Matrix {
4      int N, M;
5      vector<vector<int>>> mat;
6      Matrix(int n, int m, bool I = false) : N(n), M(m) {
7          mat.assign(N, vector<int>(M, 0ll));
8          if (I) {
9              for (int i = 0; i < min(N, M); i++) {
10                 mat[i][i] = 1;
11             }
12         }
13     };
14     Matrix operator*(const Matrix &other) {
15         assert(M == other.N);
16         Matrix res(N, other.M);
17         for (int i = 0; i < N; i++) {
18             for (int k = 0; k < M; k++) {
19                 for (int j = 0; j < other.M; j++) {
20                     res.mat[i][j] = (res.mat[i][k] + mat[i][k]
21                                     * other.mat[k][j]) % MOD;
22                 }
23             }
24         }
25         return res;
26     }
27     Matrix exp(Matrix A, int B) {
28         assert(A.N == A.M);
29         Matrix res(A.N, A.M, true);
30         while (B) {
31             if (B & 1) res = res * A;
32             A = A * A;
33             B >>= 1;
34         }
35         return res;
36     };

```

7.10 Combinatorics

```

1  const int N = 200005;
2  const int MOD = 1e9 + 7;
3  // DEFINE INT LONG LONG PLMDS
4  int fat[N], fati[N];
5
6  // (a^b) % m em O(log b)
7  // coloque o fexp
8
9  int inv(int n) { return fexp(n, MOD - 2); }
10
11 void precalc() {
12     fat[0] = 1;
13     fati[0] = 1;
14     for (int i = 1; i < N; i++) fat[i] = (fat[i - 1] * i) %
15     MOD;
16     fati[N - 1] = inv(fat[N - 1]);
17     for (int i = N - 2; i >= 0; i--) fati[i] = (fati[i + 1] *
18     (i + 1)) % MOD;
19 }
20
21 int choose(int n, int k) {
22     if (k < 0 || k > n) return 0;
23     return (((fat[n] * fati[k]) % MOD) * fati[n - k]) % MOD;
24 }
25
26 // n! / (n-k)!
27 int perm(int n, int k) {
28     if (k < 0 || k > n) return 0;
29     return (fat[n] * fati[n - k]) % MOD;
30 }
31
32 // C_n = (1 / (n+1)) * C(2n, n)
33 int catalan(int n) {
34     if (n < 0 || 2 * n >= N) return 0;
35     int c2n_n = choose(2 * n, n);
36     return (c2n_n * inv(n + 1)) % MOD;

```

7.11 Divisores

```

1  // Retorna um vetor com os divisores de x
2  // eh preciso ter o crivo implementado
3  // 0(divisores)

```

```

4
5  vector<int> divs(int x){
6      vector<int> ans = {1};
7      vector<array<int, 2>> primos; // {primo, expoente}
8
9      while (x > 1) {
10         int p = crivo[x], cnt = 0;
11         while (x % p == 0) cnt++, x /= p;
12         primos.push_back({p, cnt});
13     }
14
15     for (int i=0; i<primos.size(); i++){
16         int cur = 1, len = ans.size();
17
18         for (int j=0; j<primos[i][1]; j++){
19             cur *= primos[i][0];
20             for (int k=0; k<len; k++){
21                 ans.push_back(cur*ans[k]);
22             }
23         }
24     }
25     return ans;
26 }

```

7.12 Segment Sieve

```

1  // Retorna quantos primos tem entre [l, r] (inclusivo)
2  // precisa de um vetor com os primos ate sqrt(r)
3  int seg_sieve(int l, int r){
4      if (l > r) return 0;
5      vector<bool> is_prime(r - l + 1, true);
6      if (l == 1) is_prime[0] = false;
7
8      for (int p : primos){
9          if (p * p > r) break;
10         int start = max(p * p, (l + p - 1) / p * p);
11         for (int j = start; j <= r; j += p){
12             if (j >= l) {
13                 is_prime[j - l] = false;
14             }
15         }
16     }
17
18     return accumulate(all(is_prime), 0ll);
19 }

```

7.13 Discrete Log

```

1  // returns minimum x for which a^x = b (mod m), a and m are
2  // coprime.
3  // if the answer dont need to be greater than some value, the
4  // vector<int> can be removed
5  int discrete_log(int a, int b, int m) {
6      a %= m, b %= m;
7      int n = sqrt(m) + 1;
8
9      int an = 1;
10     for (int i = 0; i < n; ++i)
11         an = (an * 1ll * a) % m;
12
13     unordered_map<int, vector<int>> vals;
14     for (int q = 0, cur = b; q <= n; ++q) {
15         vals[cur].push_back(q);
16         cur = (cur * 1ll * a) % m;
17     }
18
19     int res = LLONG_MAX;
20
21     for (int p = 1, cur = 1; p <= n; ++p) {
22         cur = (cur * 1ll * an) % m;
23         if (vals.count(cur)) {
24             for (int q: vals[cur]){
25                 int ans = n * p - q;
26                 res = min(res, ans);
27             }
28         }
29     }
30     return res;
31 }

```

7.14 Mod Inverse

```

1  array<int, 2> extended_gcd(int a, int b) {

```

```

2     if (b == 0) return {1, 0};
3     auto [x, y] = extended_gcd(b, a % b);
4     return {y, x - (a / b) * y};
5 }
6
7 int mod_inverse(int a, int m) {
8     auto [x, y] = extended_gcd(a, m);
9     return (x % m + m) % m;
10 }

```

7.15 Fft

```

1 // multiplica dois polinomios em O(NlogN)
2
3 using cd = complex<double>;
4 const double PI = acos(-1);
5
6 void fft(vector<cd> &A, bool invert) {
7     int N = size(A);
8
9     for (int i = 1, j = 0; i < N; i++) {
10         int bit = N >> 1;
11         for (; j < bit; bit >>= 1)
12             j ^= bit;
13         j ^= bit;
14
15         if (i < j)
16             swap(A[i], A[j]);
17     }
18
19     for (int len = 2; len <= N; len <= 1) {
20         double ang = 2 * PI / len * (invert ? -1 : 1);
21         cd wlen(cos(ang), sin(ang));
22         for (int i = 0; i < N; i += len) {
23             cd w(1);
24             for (int j = 0; j < len/2; j++) {
25                 cd u = A[i+j], v = A[i+j+len/2] * w;
26                 A[i+j] = u + v;
27                 A[i+j+len/2] = u - v;
28                 w *= wlen;
29             }
30         }
31     }
32
33     if (invert) {
34         for (auto &x : A)
35             x /= N;
36     }
37 }
38
39 vector<int> multiply(vector<int> const& A, vector<int> const&
40 B) {
41     vector<cd> fa(begin(A), end(A)), fb(begin(B), end(B));
42     int N = 1;
43     while (N < size(A) + size(B))
44         N <<= 1;
45     fa.resize(N);
46     fb.resize(N);
47
48     fft(fa, false);
49     fft(fb, false);
50     for (int i = 0; i < N; i++)
51         fa[i] *= fb[i];
52     fft(fa, true);
53
54     vector<int> result(N);
55     for (int i = 0; i < N; i++)
56         result[i] = round(fa[i].real());
57     return result;
58 }

```

8 Primitives

9 Graph

9.1 Min Cost Max Flow

```

1 // Encontra o menor custo para passar K de fluxo em um grafo
2 // com N vertices
3 // Funciona com multiplas arestas para o mesmo par de
4 // vertices

```

```

3 // Para encontrar o min cost max flow eh so fazer K =
4 // infinito
5 struct Edge {
6     int from, to, capacity, cost, id;
7 };
8
9 vector<vector<array<int, 2>>> adj;
10 vector<Edge> edges; // arestas pares sao as normais e suas
11 // reversas sao as impares
12
13 const int INF = LLONG_MAX;
14
15 void shortest_paths(int n, int v0, vector<int> &dist, vector<
16 int> &edge_to) {
17     dist.assign(n, INF);
18     dist[v0] = 0;
19     vector<bool> in_queue(n, false);
20     queue<int> q;
21     q.push(v0);
22     edge_to.assign(n, -1);
23
24     while (!q.empty()) {
25         int u = q.front();
26         q.pop();
27         in_queue[u] = false;
28         for (auto [v, id] : adj[u]) {
29             if (edges[id].capacity > 0 && dist[v] > dist[u] +
30 edges[id].cost) {
31                 dist[v] = dist[u] + edges[id].cost;
32                 edge_to[v] = id;
33                 if (!in_queue[v]) {
34                     in_queue[v] = true;
35                     q.push(v);
36                 }
37             }
38         }
39     }
40 }
41
42 void add_edge(int from, int to, int capacity, int cost) {
43     edges.push_back({from, to, capacity, cost, (int)edges.
44 size()});
45     edges.push_back({to, from, 0, -cost, (int)edges.size()});
46     // reversa
47 }
48
49 int min_cost_flow(int N, int K, int s, int t) {
50     adj.assign(N, vector<array<int, 2>>());
51
52     for (Edge e : edges) {
53         adj[e.from].push_back({e.to, e.id});
54     }
55
56     int flow = 0;
57     int cost = 0;
58     vector<int> dist, edge_to;
59     while (flow < K) {
60         shortest_paths(N, s, dist, edge_to);
61         if (dist[t] == INF)
62             break;
63
64         // find max flow on that path
65         int f = K - flow;
66         int cur = t;
67         while (cur != s) {
68             f = min(f, edges[edge_to[cur]].capacity);
69             cur = edges[edge_to[cur]].from;
70         }
71
72         // apply flow
73         flow += f;
74         cost += f * dist[t];
75         cur = t;
76         while (cur != s) {
77             int edge = edge_to[cur];
78             int rev_edge = edge^1;
79
80             edges[edge].capacity -= f;
81             edges[rev_edge].capacity += f;
82             cur = edges[edge].from;
83         }
84     }
85
86     if (flow < K)
87         return -1;
88 }

```

```

83     else
84         return cost;
85 }

```

9.2 Floyd Warshall

```

1 // SSP e acha ciclos.
2 // Bom com constraints menores.
3 // O(n^3)
4
5 int dist[501][501];
6
7 void floydWarshall() {
8     for(int k = 0; k < n; k++) {
9         for(int i = 0; i < n; i++) {
10            for(int j = 0; j < n; j++) {
11                dist[i][j] = min(dist[i][j], dist[i][k] +
12                                dist[k][j]);
13            }
14        }
15    }
16    void solve() {
17        int m, q;
18        cin >> n >> m >> q;
19        for(int i = 0; i < n; i++) {
20            for(int j = i; j < n; j++) {
21                if(i == j) {
22                    dist[i][j] = dist[j][i] = 0;
23                } else {
24                    dist[i][j] = dist[j][i] = linf;
25                }
26            }
27        }
28        for(int i = 0; i < m; i++) {
29            int u, v, w;
30            cin >> u >> v >> w; u--; v--;
31            dist[u][v] = min(dist[u][v], w);
32            dist[v][u] = min(dist[v][u], w);
33        }
34        floydWarshall();
35        while(q--) {
36            int u, v;
37            cin >> u >> v; u--; v--;
38            if(dist[u][v] == linf) cout << -1 << '\n';
39            else cout << dist[u][v] << '\n';
40        }
41    }

```

9.3 Edmonds-karp

```

1 // Edmonds-Karp com scalling O(EÅlog(F))
2
3 int n, m;
4 const int MAXN = 510;
5 vector<vector<int>> capacity(MAXN, vector<int>(MAXN, 0));
6 vector<vector<int>> adj(MAXN);
7
8 int bfs(int s, int t, int scale, vector<int>& parent) {
9     fill(parent.begin(), parent.end(), -1);
10    parent[s] = -2;
11    queue<pair<int, int>> q;
12    q.push({s, LLONG_MAX});
13
14    while (!q.empty()) {
15        int cur = q.front().first;
16        int flow = q.front().second;
17        q.pop();
18
19        for (int next : adj[cur]) {
20            if (parent[next] == -1 && capacity[cur][next] >=
21                scale) {
22                parent[next] = cur;
23                int new_flow = min(flow, capacity[cur][next]);
24                q.push({next, new_flow});
25            }
26        }
27    }
28    return 0;
29
30    return 0;
31 }

```

```

32
33 int maxflow(int s, int t) {
34     int flow = 0;
35     vector<int> parent(MAXN);
36     int new_flow;
37     int scalling = 1ll << 62;
38
39     while (scalling > 0) {
40         while (new_flow = bfs(s, t, scalling, parent)){
41             if (new_flow == 0) continue;
42             flow += new_flow;
43             int cur = t;
44             while (cur != s) {
45                 int prev = parent[cur];
46                 capacity[prev][cur] -= new_flow;
47                 capacity[cur][prev] += new_flow;
48                 cur = prev;
49             }
50         }
51         scalling /= 2;
52     }
53
54     return flow;
55 }

```

9.4 Lca Jc

```

1 const int MAXN = 200005;
2 int N;
3 int LOG;
4
5 vector<vector<int>> adj;
6 vector<int> profundidade;
7 vector<vector<int>> cima; // cima[v][j] eh o 2^j-esimo
8                          // ancestral de v
9
10 void dfs(int v, int p, int d) {
11     profundidade[v] = d;
12     cima[v][0] = p; // o pai direto eh o 2^0-Åximo ancestral
13     for (int j = 1; j < LOG; j++) {
14         // se o ancestral 2^(j-1) existir, calculamos o 2^j
15         if (cima[v][j-1] != -1) {
16             cima[v][j] = cima[cima[v][j-1]][j-1];
17         } else {
18             cima[v][j] = -1; // nao tem ancestral superior
19         }
20     }
21     for (int nei : adj[v]) {
22         if (nei != p) {
23             dfs(nei, v, d + 1);
24         }
25     }
26 }
27
28 void build(int root) {
29     LOG = ceil(log2(N));
30     profundidade.assign(N + 1, 0);
31     cima.assign(N + 1, vector<int>(LOG, -1));
32     dfs(root, -1, 0);
33 }
34
35 int get_lca(int a, int b) {
36     if (profundidade[a] < profundidade[b]) {
37         swap(a, b);
38     }
39     // sobe 'a' ate a mesma profundidade de 'b'
40     for (int j = LOG - 1; j >= 0; j--) {
41         if (profundidade[a] - (1 << j) >= profundidade[b]) {
42             a = cima[a][j];
43         }
44     }
45     // se 'b' era um ancestral de 'a', entÅo 'a' agora Å
46     // igual a 'b'
47     if (a == b) {
48         return a;
49     }
50     // sobe os dois juntos ate encontrar os filhos do LCA
51     for (int j = LOG - 1; j >= 0; j--) {
52         if (cima[a][j] != -1 && cima[a][j] != cima[b][j]) {
53             a = cima[a][j];
54             b = cima[b][j];
55         }
56     }
57     return cima[a][0];
58 }

```

9.5 Diametro

```

1  /*
2  0(N + M), retorna cara mais distante, len
3  do caminho em arestas e os caras do caminho em um vetor
4  Para pegar o diametro da arvore, chamar duas vezes
5  auto [a, DA, PA] = calc(1);
6  auto [b, len_diametro, caminho_diametro] = calc(a);
7  */
8  auto calc = [&](int S) -> tuple<int, int, vector<int>> {
9      queue<pair<int, int>> q;
10     q.push({S, 0});
11     vector<int> dp(n + 1, -1), pai(n + 1, 0);
12     dp[S] = 0;
13     pai[S] = -1;
14     int longe = S;
15     int dist = 0;
16     while (!q.empty()){
17         auto [u, d] = q.front();
18         q.pop();
19         if (d > dist){
20             dist = d;
21             longe = u;
22         }
23         for (int v : adj[u]){
24             if (dp[v] == -1){
25                 dp[v] = d + 1;
26                 pai[v] = u;
27                 q.push({v, d + 1});
28             }
29         }
30     }
31     int cur = longe;
32     vector<int> caminho;
33     while (cur != -1){
34         caminho.push_back(cur);
35         cur = pai[cur];
36     }
37     return { longe, dist, caminho };
38 };

```

9.6 Eulerian Path

```

1  /**
2  * versao que assume: #define int long long
3  *
4  * Retorna um caminho/ciclo euleriano em um grafo (se existir
5  * - g: lista de adjacencia (vector<vector<int>>)).
6  * - directed: true se o grafo for dirigido.
7  * - s: vertice inicial.
8  * - e: vertice final (opcional). Se informado, tenta caminho
9  *   de s ate e.
10 * - 0(Nlog(N))
11 * Retorna vetor com a sequencia de vertices, ou vazio se
12 *   impossivel.
13 */
14 vector<int> eulerian_path(const vector<vector<int>>& g, bool
15     directed, int s, int e = -1) {
16     int n = (int)g.size();
17     // copia das adjacencias em multiset para permitir
18     // remoção especifica
19     vector<multiset<int>> h(n);
20     vector<int> in_degree(n, 0);
21     vector<int> result;
22     stack<int> st;
23     // preencher h e indegrees
24     for (int u = 0; u < n; ++u) {
25         for (auto v : g[u]) {
26             ++in_degree[v];
27             h[u].emplace(v);
28         }
29     }
30     st.emplace(s);
31     if (e != -1) {
32         int out_s = (int)h[s].size();
33         int out_e = (int)h[e].size();
34         int diff_s = in_degree[s] - out_s;
35         int diff_e = in_degree[e] - out_e;
36         if (diff_s * diff_e != -1) return {}; // impossivel
37     }
38     for (int u = 0; u < n; ++u) {
39         if (e != -1 && (u == s || u == e)) continue;
40         int out_u = (int)h[u].size();

```

```

37         if (in_degree[u] != out_u || (!directed && (in_degree
38             [u] & 1))) {
39             return {};
40         }
41     }
42     while (!st.empty()) {
43         int u = st.top();
44         if (h[u].empty()) {
45             result.emplace_back(u);
46             st.pop();
47         } else {
48             int v = *h[u].begin();
49             auto it = h[u].find(v);
50             if (it != h[u].end()) h[u].erase(it);
51             --in_degree[v];
52             if (!directed) {
53                 auto it2 = h[v].find(u);
54                 if (it2 != h[v].end()) h[v].erase(it2);
55                 --in_degree[u];
56             }
57             st.emplace(v);
58         }
59     }
60     for (int u = 0; u < n; ++u) {
61         if (in_degree[u] != 0) return {};
62     }
63     reverse(result.begin(), result.end());
64     return result;

```

9.7 Lca

```

1  // LCA - CP algorithm
2  // preprocessing 0(NlogN)
3  // lca 0(logN)
4  // Uso: criar LCA com a quantidade de vertices (n) e lista de
5  //   adjacencias (adj)
6  // chamar a funcao preprocess com a raiz da arvore
7  struct LCA {
8      int n, l, timer;
9      vector<vector<int>> adj;
10     vector<int> tin, tout;
11     vector<vector<int>> up;
12
13     LCA(int n, const vector<vector<int>>& adj) : n(n), adj(
14         adj) {}
15
16     void dfs(int v, int p) {
17         tin[v] = ++timer;
18         up[v][0] = p;
19         for (int i = 1; i <= l; ++i)
20             up[v][i] = up[up[v][i-1]][i-1];
21
22         for (int u : adj[v]) {
23             if (u != p)
24                 dfs(u, v);
25         }
26         tout[v] = ++timer;
27     }
28
29     bool is_ancestor(int u, int v) {
30         return tin[u] <= tin[v] && tout[u] >= tout[v];
31     }
32
33     int lca(int u, int v) {
34         if (is_ancestor(u, v))
35             return u;
36         if (is_ancestor(v, u))
37             return v;
38         for (int i = l; i >= 0; --i) {
39             if (!is_ancestor(up[u][i], v))
40                 u = up[u][i];
41         }
42         return up[u][0];
43     }
44
45     void preprocess(int root) {
46         tin.resize(n);
47         tout.resize(n);
48         timer = 0;
49         l = ceil(log2(n));
50         up.assign(n, vector<int>(l + 1));
51         dfs(root, root);
52     }

```

```
53 };
```

9.8 Topological Sort

```
1 vector<int> adj[MAXN];
2 vector<int> estado(MAXN); // 0: nao visitado 1: processamento
3 // 2: processado
4 vector<int> ordem;
5 bool temCiclo = false;
6
7 void dfs(int v) {
8     if(estado[v] == 1) {
9         temCiclo = true;
10        return;
11    }
12    if(estado[v] == 2) return;
13    estado[v] = 1;
14    for(auto &nei : adj[v]) {
15        if(estado[nei] != 2) dfs(nei);
16    }
17    estado[v] = 2;
18    ordem.push_back(v);
19    return;
20 }
```

9.9 Dijkstra

```
1 // SSP com pesos positivos.
2 // O((V + E) log V).
3
4 vector<int> dijkstra(int S) {
5     vector<bool> vis(MAXN, 0);
6     vector<ll> dist(MAXN, LLONG_MAX);
7     dist[S] = 0;
8     priority_queue<pii, vector<pii>, greater<pii>> pq;
9     pq.push({0, S});
10    while(pq.size()) {
11        ll v = pq.top().second;
12        pq.pop();
13        if(vis[v]) continue;
14        vis[v] = 1;
15        for(auto &[peso, vizinho] : adj[v]) {
16            if(dist[vizinho] > dist[v] + peso) {
17                dist[vizinho] = dist[v] + peso;
18                pq.push({dist[vizinho], vizinho});
19            }
20        }
21    }
22    return dist;
23 }
```

9.10 Acha Pontes

```
1 vector<int> d, low, pai; // d[v] Tempo de descoberta (
2 // discovery time)
3 vector<bool> vis;
4 vector<int> pontos_articulacao;
5 vector<pair<int, int>> pontes;
6 int tempo;
7
8 vector<vector<int>> adj;
9
10 void dfs(int u) {
11     vis[u] = true;
12     tempo++;
13     d[u] = low[u] = tempo;
14     int filhos_dfs = 0;
15     for (int v : adj[u]) {
16         if (v == pai[u]) continue;
17         if (vis[v]) { // back edge
18             low[u] = min(low[u], d[v]);
19         } else {
20             pai[v] = u;
21             filhos_dfs++;
22             dfs(v);
23             low[u] = min(low[u], low[v]);
24             if (pai[u] == -1 && filhos_dfs > 1) {
25                 pontos_articulacao.push_back(u);
26             }
27             if (pai[u] != -1 && low[v] >= d[u]) {
28                 pontos_articulacao.push_back(u);
29             }
30             if (low[v] > d[u]) {
```

```
30         pontes.push_back({min(u, v), max(u, v)});
31     }
32 }
33 }
34 }
```

9.11 Khan

```
1 // topo-sort DAG
2 // lexicograficamente menor.
3 // N: numero de vertices (1-indexado)
4 // adj: lista de adjacencia do grafo
5
6 const int MAXN = 5 * 1e5 + 2;
7 vector<int> adj[MAXN];
8 int N;
9
10 vector<int> kahn() {
11     vector<int> indegree(N + 1, 0);
12     for (int u = 1; u <= N; u++) {
13         for (int v : adj[u]) {
14             indegree[v]++;
15         }
16     }
17     priority_queue<int, vector<int>, greater<int>> pq;
18     for (int i = 1; i <= N; i++) {
19         if (indegree[i] == 0) {
20             pq.push(i);
21         }
22     }
23     vector<int> result;
24     while (!pq.empty()) {
25         int u = pq.top();
26         pq.pop();
27         result.push_back(u);
28         for (int v : adj[u]) {
29             indegree[v]--;
30             if (indegree[v] == 0) {
31                 pq.push(v);
32             }
33         }
34     }
35     if (result.size() != N) {
36         return {};
37     }
38     return result;
39 }
```

9.12 Bellman Ford

```
1 struct Edge {
2     int u, v, w;
3 };
4
5 // se x = -1, nao tem ciclo
6 // se x != -1, pegar pais de x pra formar o ciclo
7
8 int n, m;
9 vector<Edge> edges;
10 vector<int> dist(n);
11 vector<int> pai(n, -1);
12
13 for (int i = 0; i < n; i++) {
14     x = -1;
15     for (Edge &e : edges) {
16         if (dist[e.u] + e.w < dist[e.v]) {
17             dist[e.v] = max(-INF, dist[e.u] + e.w);
18             pai[e.v] = e.u;
19             x = e.v;
20         }
21     }
22 }
23
24 // achando caminho (se precisar)
25 for (int i = 0; i < n; i++) x = pai[x];
26
27 vector<int> ciclo;
28 for (int v = x; v = pai[v]) {
29     ciclo.push_back(v);
30     if (v == x && ciclo.size() > 1) break;
31 }
32 reverse(ciclo.begin(), ciclo.end());
```

9.13 Dinitz

```

1 // Complexidade:  $O(V^2E)$ 
2
3 struct FlowEdge {
4     int from, to;
5     long long cap, flow = 0;
6     FlowEdge(int from, int to, long long cap) : from(from),
7         to(to), cap(cap) {}
8 };
9
10 struct Dinic {
11     const long long flow_inf = 1e18;
12     vector<FlowEdge> edges;
13     vector<vector<int>> adj;
14     int qntV, qntE = 0;
15     int source, sink;
16     vector<int> level, ptr;
17     queue<int> q;
18
19     Dinic(int qntV, int source, int sink) : qntV(qntV),
20         source(source), sink(sink) {
21         adj.resize(qntV);
22         level.resize(qntV);
23         ptr.resize(qntV);
24     }
25
26     void add_edge(int from, int to, long long cap) {
27         edges.emplace_back(from, to, cap);
28         edges.emplace_back(to, from, 0);
29         adj[from].push_back(qntE);
30         adj[to].push_back(qntE + 1);
31         qntE += 2;
32     }
33
34     bool bfs() {
35         while (!q.empty()) {
36             int from = q.front();
37             q.pop();
38             for (int id : adj[from]) {
39                 if (edges[id].cap == edges[id].flow)
40                     continue;
41                 if (level[edges[id].to] != -1)
42                     continue;
43                 level[edges[id].to] = level[from] + 1;
44                 q.push(edges[id].to);
45             }
46         }
47         return level[sink] != -1;
48     }
49
50     long long dfs(int from, long long pushed) {
51         if (pushed == 0)
52             return 0;
53         if (from == sink)
54             return pushed;
55         for (int& cid = ptr[from]; cid < (int)adj[from].size
56             (); cid++) {
57             int id = adj[from][cid];
58             int to = edges[id].to;
59             if (level[from] + 1 != level[to])
60                 continue;
61             long long tr = dfs(to, min(pushed, edges[id].cap
62                 - edges[id].flow));
63             if (tr == 0)
64                 continue;
65             edges[id].flow += tr;
66             edges[id ^ 1].flow -= tr;
67             return tr;
68         }
69         return 0;
70     }
71
72     long long max_flow() {
73         long long f = 0;
74         while (true) {
75             fill(level.begin(), level.end(), -1);
76             level[source] = 0;
77             q.push(source);
78             if (!bfs())
79                 break;
80             fill(ptr.begin(), ptr.end(), 0);
81             while (long long pushed = dfs(source, flow_inf))
82                 f += pushed;
83         }
84     }
85
86     return f;
87 }

```

```

82     }
83 };

```

9.14 Kruskal

```

1 // Ordena as arestas por peso, insere se ja nao estiver no
2 // mesmo componente
3 //  $O(E \log E)$ 
4
5 struct Edge {
6     int u, v, w;
7     bool operator <(Edge const & other) {
8         return weight < other.weight;
9     }
10 }
11
12 vector<Edge> kruskal(int n, vector<Edge> edges) {
13     vector<Edge> mst;
14     DSU dsu = DSU(n + 1);
15     sort(edges.begin(), edges.end());
16     for (Edge e : edges) {
17         if (dsu.find(e.u) != dsu.find(e.v)) {
18             mst.push_back(e);
19             dsu.join(e.u, e.v);
20         }
21     }
22     return mst;
23 }

```

9.15 Kosaraju

```

1 bool vis[MAXN];
2 vector<int> order;
3 int component[MAXN];
4 int N, m;
5 vector<int> adj[MAXN], adj_rev[MAXN];
6
7 // dfs no grafo original para obter a ordem (pos-order)
8 void dfs1(int u) {
9     vis[u] = true;
10    for (int v : adj[u]) {
11        if (!vis[v]) {
12            dfs1(v);
13        }
14    }
15    order.push_back(u);
16 }
17
18 // dfs o grafo reverso para encontrar os SCCs
19 void dfs2(int u, int c) {
20     component[u] = c;
21     for (int v : adj_rev[u]) {
22         if (component[v] == -1) {
23             dfs2(v, c);
24         }
25     }
26 }
27
28 int kosaraju() {
29     order.clear();
30     fill(vis + 1, vis + N + 1, false);
31     for (int i = 1; i <= N; i++) {
32         if (!vis[i]) {
33             dfs1(i);
34         }
35     }
36     fill(component + 1, component + N + 1, -1);
37     int c = 0;
38     reverse(order.begin(), order.end());
39     for (int u : order) {
40         if (component[u] == -1) {
41             dfs2(u, c++);
42         }
43     }
44     return c;
45 }

```

10 DP

10.1 Bitmask

```

1 // dp de intervalos com bitmask

```

```

2 int prox(int idx) {
3     return lower_bound(S.begin(), S.end(), array<int, 4>{S[
4         idx][1], 011, 011, 011}) - S.begin();
5 }
6 int dp[1002][1000] = {0};
7
8 int rec(int i, int vis) {
9     if (i == (int)S.size()) {
10         if (__builtin_popcountll(vis) == N) return 0;
11         return LLONG_MIN;
12     }
13     if (dp[i][vis] != -1) return dp[i][vis];
14     int ans = rec(i + 1, vis);
15     ans = max(ans, rec(prox(i), vis | (1ll << S[i][3])) + S[i
16         ][2]);
17     return dp[i][vis] = ans;
18 }

```

10.2 Kadane

```

1 // O(n) - retorna maximum subarray sum
2 array<int, 3> kadane(vector<int> &a) {
3     int L = -1, R = -1, n = a.size();
4     int soma = 0, ans = -INF;
5     int fim = n - 1;
6     for (int i = n - 1; i >= 0; i--) {
7         if (a[i] > soma) {
8             soma = a[i];
9             fim = i;
10        } else {
11            soma += a[i];
12        }
13        if (soma > ans) {
14            ans = soma;
15            L = i;
16            R = fim;
17        }
18    }
19    return {ans, L, R};
20 }

```

10.3 Knapsack

```

1 // dp[i][j] => i-esimo item com j-carga sobrando na mochila
2 // O(N * W)
3 vector<vector<int>> dp(N + 1, vector<int>(W + 1, 0));
4 for (int i = 1; i <= N; i++) dp[i][0] = 0;
5 for (int j = 0; j <= W; j++) dp[0][j] = 0;
6 for (int i = 1; i <= N; i++) {
7     for (int j = 0; j <= W; j++) {
8         dp[i][j] = dp[i - 1][j];
9         if (j >= items[i - 1].first) {
10            dp[i][j] = max(dp[i][j], dp[i - 1][j - items[i
11                - 1].first] + items[i - 1].second);
12        }
13    }
14 }
15 cout << dp[N][W] << '\n';

```

10.4 Digit

```

1 vector<int> digits;
2
3 int dp[20][10][2][2];
4
5 int rec(int i, int last, int flag, int started) {
6     if (i == (int)digits.size()) return 1;
7     if (dp[i][last][flag][started] != -1) return dp[i][last][
8         flag][started];
9     int lim;
10    if (flag) lim = 9;
11    else lim = digits[i];
12    int ans = 0;
13    for (int d = 0; d <= lim; d++) {
14        if (started && d == last) continue;
15        int new_flag = flag;
16        int new_started = started;
17        if (d > 0) new_started = 1;
18        if (!flag && d < lim) new_flag = 1;
19        ans += rec(i + 1, d, new_flag, new_started);
20    }
21    return dp[i][last][flag][started] = ans;
22 }

```

10.5 Edit Distance

```

1 vector<vector<int>> dp(n+1, vector<int>(m+1, LLONG_MIN));
2 for (int j = 0; j <= m; j++) dp[0][j] = j;
3 for (int i = 0; i <= n; i++) dp[i][0] = i;
4 for (int i = 1; i <= n; i++) {
5     for (int j = 1; j <= m; j++) {
6         if (a[i-1] == b[j-1]) {
7             dp[i][j] = dp[i-1][j-1];
8         } else {
9             dp[i][j] = min({dp[i-1][j] + 1, dp[i][j-1] +
10                 1, dp[i-1][j-1] + 1});
11         }
12     }
13 }
14 cout << dp[n][m];

```

10.6 Lis Brunomonteiro

```

1 // Calcula e retorna uma LIS O(n.log(n))
2 template<typename T> vector<T> lis(vector<T> &v) {
3     int n = v.size(), m = -1;
4     vector<T> d(n+1, INF);
5     vector<int> l(n);
6     d[0] = -INF;
7
8     for (int i = 0; i < n; i++) {
9         // Para non-decreasing use upper_bound()
10        int t = lower_bound(d.begin(), d.end(), v[i]) - d.
11            begin();
12        d[t] = v[i], l[i] = t, m = max(m, t);
13    }
14
15    int p = n;
16    vector<T> ret;
17    while (p-- > 0) {
18        if (l[p] == m) {
19            ret.push_back(v[p]);
20            m--;
21        }
22    }
23    reverse(ret.begin(), ret.end());
24    return ret;
25 }

```

10.7 Disjoint Blocks

```

1 // num max de subarrays disjuntos com soma x usando apenas
2 // prefixo i (ou seja, considerando prefixo a[1..i]).
3 int disjointSumX(vector<int> &a, int x) {
4     int n = a.size();
5     map<int, int> best; // best[pref] = melhor dp visto para
6     esse pref
7     best[0] = 0;
8     int pref = 0;
9     vector<int> dp(n + 1, 0); // dp[0] = 0
10    for (int i = 1; i <= n; i++) {
11        pref += a[i - 1];
12        dp[i] = dp[i - 1];
13        auto it = best.find(pref - x);
14        if (it != best.end()) {
15            dp[i] = max(dp[i], it->second + 1);
16        }
17        best[pref] = max(best[pref], dp[i]);
18    }
19    return dp[n];
20 }

```

10.8 Lcs

```

1 string s1, s2;
2 int dp[1001][1001];
3 int lcs(int i, int j) {
4     if (i < 0 || j < 0) return 0;
5     if (dp[i][j] != -1) return dp[i][j];
6     if (s1[i] == s2[j]) return dp[i][j] = 1 + lcs(i - 1, j -
7         1);
8     else return dp[i][j] = max(lcs(i - 1, j), lcs(i, j - 1));
9 }

```

10.9 Lis

```

1 int lis_nlogn(vector<int> &v) {
2     vector<int> lis;
3     lis.push_back(v[0]);
4     for (int i = 1; i < v.size(); i++) {
5         if (v[i] > lis.back()) {
6             // estende a lis
7             lis.push_back(v[i]);
8         } else {
9             // encontra o primeiro elemento em lis que >= v[i]
10            auto it = lower_bound(lis.begin(), lis.end(), v[i]);
11            *it = v[i];
12        }
13    }
14    return lis.size();
15 }
16
17 // lis na tree do problema da sub
18 const int MAXN_TREE = 100001;
19 vector<int> adj[MAXN_TREE];
20 int values[MAXN_TREE];
21 int ans = 0;
22
23 void dfs(int u, int p, vector<int> &tails) {
24     auto it = lower_bound(tails.begin(), tails.end(), values[u]);
25     int prev = -1;
26     bool coloquei = false;
27     if (it == tails.end()) {
28         tails.push_back(values[u]);
29         coloquei = true;
30     } else {
31         prev = *it;
32         *it = values[u];
33     }

```

```

34     ans = max(ans, (int)tails.size());
35     for (int v : adj[u]) {
36         if (v != p) {
37             dfs(v, u, tails);
38         }
39     }
40     if (coloquei) {
41         tails.pop_back();
42     } else {
43         *it = prev;
44     }
45 }

```

10.10 Lis Seg

```

1 // comprime coordenadas pra quando ai <= 1e9
2 vector<int> vals(a.begin(), a.end());
3 sort(vals.begin(), vals.end());
4 vals.erase(unique(vals.begin(), vals.end()), vals.end());
5 auto id = [&](int x) -> int {
6     return lower_bound(vals.begin(), vals.end(), x) - vals.begin();
7 };
8 for (int i = 0; i < n; i++) a[i] = id(a[i]);
9 // fim da parte de compr
10 SegTree seg(n);
11 // dp[i]: lis que termina em i
12 vector<int> dp(n, 1); // dp[i] = 1 base case
13 for (int i = 0; i < n; i++) {
14     if (a[i] > 0) dp[i] = seg.query(0, max(0ll, a[i] - 1)) + 1;
15     seg.update(a[i], dp[i]);
16 }
17 cout << *max_element(dp.begin(), dp.end()) << '\n';

```